Together First, Apart Later? When Optimizers Start to Learn Different Functions

Initial checkpoint, MIT 6.7910, Fall 2026

Juntang Wang and Víctor Conchello · November 1, 2026

Summary. We train one transformer from one initialization on one sequence of mini-batches with AdamW and with Muon (Jordan et al., 2024). (1) Muon is ahead at every step, but at equal training loss its predictions are as close to AdamW's as two data orders of one optimizer; the endpoints' loss barrier exceeds the data-order ones on one of three datasets. (2) What differs along the way, so far: how each optimizer sits at its stability threshold, how its steps align with the population gradient, how far and how evenly it moves the weights. (3) Swapping the two mid-training leaves the function within the floor of both, while the weights keep part of the parent's geometry.

Setup. The main setting is TinyStories (Eldan and Li, 2023) with a 14M-parameter nanoGPT (Karpathy, 2022) (6 layers, width 384, 4096-token BPE, 10,000 steps of 128×256 tokens, warmup over 200 steps, cosine decay to 10%); a 30M model on TinyStories and a character-level model on Shakespeare are checks (Appendix A). Muon orthogonalizes the update of each matrix in the transformer blocks and leaves embeddings, head and gains to an auxiliary AdamW; both optimizers use the best rate of a five-rate sweep. All runs share one initialization; a data order fixes the mini-batch at every step, and each optimizer runs with five. We store the next-token distributions on 2,048 fixed held-out positions; the distance between two runs is the mean Jensen–Shannon divergence (JS, bits) of these predictions. We compare runs at equal *training loss* (on 64 fixed training sequences), so that speed does not count as a difference, and against a *floor*: the distance between two data orders of one optimizer (the larger of the two means). So far 130 runs and 34 GPU-hours, with 27 more running (Appendix C).

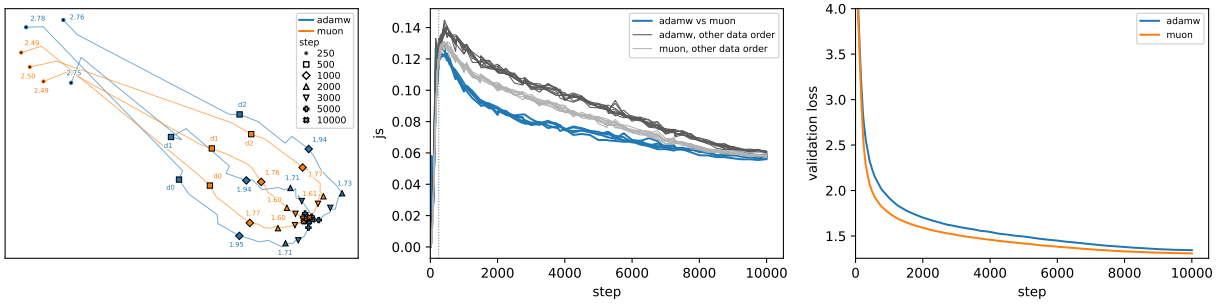


Figure 1: TinyStories-10M, AdamW (blue) and Muon (orange). Left: runs on data orders d0–d2 from step 250 on, embedded with densMAP from the distances $\sqrt{\text{JS}}$ between all logged steps (each run thinned to points at least 0.28 apart); numbers: training loss at steps 250, 1000, 2000. Middle: JS between AdamW and Muon on the same data order at equal steps (blue, five orders) and between two data orders of one optimizer (grey); dotted: step 250. Right: validation loss, mean and range over five data orders.

Step 1: how the trajectories differ (Figs. 1 and 2). After warmup, AdamW and Muon on the same data order stay close: in the embedding each such pair are nearest neighbours, and at equal steps their distance stays below that between two data orders of one optimizer. Muon is ahead: its validation loss is lower at every step (2.46 against 2.73 at step 250, 1.75 against 1.92 at step 1000; means over five data orders), and it reaches AdamW's final value, 1.344, by step 7500 of 10,000. At equal training loss (Fig. 2, left), the distance between them follows the data-order floors after warmup: below loss 4.85 it is 0.95–1.38 times the floor (median 1.02), and at the lowest loss all runs reach, 0.99 times (TinyStories-30M and Shakespeare: 1.02). In function space the two optimizers reach the same endpoint, up to the noise of the data order.

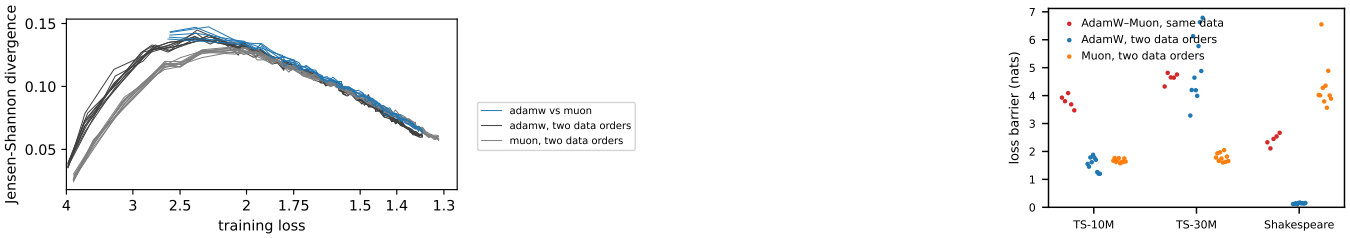


Figure 2: Left: TinyStories-10M, JS between AdamW and Muon at equal training loss L (blue, one line per data order) and between two data orders of AdamW (black) or Muon (grey); log scale in $L - 1$. Right: loss barrier between the final weights of AdamW and Muon on one data order (5 pairs) and of two data orders of one optimizer (10 pairs each).

In weight space the endpoints differ. On the straight line between two final weights, the training loss rises above the

line between its end values by a *barrier* (maximum over 11 points; Frankle et al., 2020). Between AdamW and Muon on one data order it is 3.5–4.1 nats on TinyStories-10M (mean 3.79), against 1.20–1.88 (mean 1.54) between two data orders of AdamW and 1.58–1.77 (mean 1.67) of Muon (Fig. 2, right). Only there is the change of optimizer clearly the larger one: on TinyStories-30M the AdamW–Muon barrier (mean 4.64) is above Muon's data-order barriers (mean 1.79) but below AdamW's (mean 5.05, range 3.3–6.8), and on Shakespeare (2.42) above AdamW's (0.14) but below Muon's (4.34). No two endpoints are linearly connected.

Step 2: what differs along the way (Table 1). If the functions agree at equal loss, the optimizers differ in how they get there. To see what is specific to each geometry we add, in this step only, SGD with heavy-ball momentum 0.9 at rate 0.06 (the only rate of 0.03–0.12 that never blew up on three data orders). Each optimizer sits at its own stability threshold very differently: AdamW's preconditioned sharpness for most of training, SGD's top Hessian eigenvalue only around the end of warmup, and Muon's spectral-norm sharpness at most once, in warmup. Along the step each one takes, the second-order loss change on fresh data nearly cancels the first-order one for all three ($\bar{R} \approx 1$); the adaptive steps are several times less aligned with the population gradient (ρ) and sharper on their own batch (κ). At the end, Muon has moved the weights twice as far as AdamW and ten times as far as SGD, with the most even singular values. To rank such quantities as the functions are ranked, at equal training loss and against their spread over data orders, we are measuring about 30 whole-model metrics on nine new runs (three optimizers, three data orders) with checkpoints at consecutive steps.

Table 1: TinyStories-10M. Thresholds: $\lambda_{\max}(H) < (2+2\mu)/\eta_t$ (SGD); $\lambda_{\max}(P^{-1}H) < (2+2\beta_1)/((1-\beta_1)\eta_t)$, P AdamW's preconditioner (Cohen et al., 2022); spectral-norm sharpness $S < 2\|\nabla L\|_*/\eta_t$ (Muon; Islamov et al., 2026). At threshold: share of measured steps at ≥ 0.9 of it (Muon: S at most 0.93 of the bound, in warmup; 0.39 at step 3000). Second half of 3000 steps at a constant rate (3 rates, d0): $R = s^\top Hs/(-2g^\top s)$ along the step s, with g, H on fresh data ($\bar{R}$) or the step's batch (R_B); $R_B = \bar{R}\kappa\rho$, $\rho = \bar{g}^\top s/g_B^\top s$, $\kappa = s^\top H_B s/s^\top \bar{H} s$. Weights at step 10,000; stable rank $\|W\|_F^2/\|W\|_2^2$, mean over block matrices.

	SGD	AdamW	Muon
At threshold	6–10% (steps 221–574 of d0)	50–65% (up to step 4500–9500)	step 50 of d0
$\bar{R}$, fresh data; R_B, own batch	0.90–1.05; 0.34–0.67	1.01–1.03; 0.13–0.15	0.95–1.00; 0.08–0.10
coherence ρ; curvature ratio κ	0.30–0.66; 1.00–1.05	0.10–0.12; up to 1.26	0.06–0.07; up to 1.4
$\|\theta-\theta_0\|/\|\theta_0\|$; stable rank	0.53; 56	2.72; 41	5.04; 75

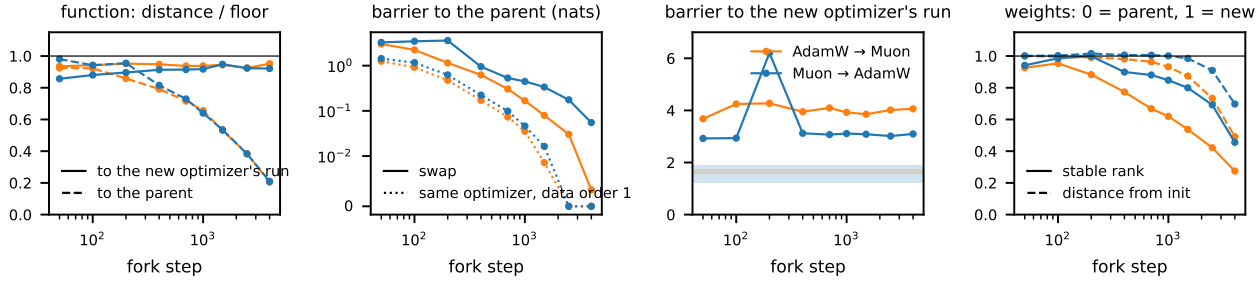


Figure 3: Swaps between AdamW and Muon on TinyStories-10M (data order 0), final weights of each fork against the fork step. Left: JS to the new optimizer's pure run and to the parent, each over that run's data-order floor. Second: barrier to the parent, for the swap and for the parent's optimizer continued on data order 1. Third: barrier to the new optimizer's pure run; bands: its data-order barriers. Right: stable rank and distance from the initialization, as a fraction of the way from the parent's value (0) to the new optimizer's (1).

Step 3: swaps (Fig. 3). We switch between AdamW and Muon at nine fork steps from 50 to 4000, with a warm optimizer state and the parent's batches (Appendix B). Every fork ends within the data-order floor of the new optimizer's pure run (0.86–0.95 floors) and, increasingly with the fork step, of the parent's (0.92–0.98 at step 50, 0.20–0.21 at step 4000): as the pure runs end within each other's floor, the function cannot tell which optimizer decided it. The weights can. A fork stays linearly connected to its parent (barrier below 0.05 nats) only from a later fork step than a change of data order: from AdamW at step 2500 (0.03) against 1000 (0.035), from Muon not even at step 4000 (0.055) against 1000 (0.046). No fork ends in the basin of the new optimizer's pure run (barriers 3.7–4.3 nats into Muon and 2.9–6.2 into AdamW, against data-order barriers of 1.6–1.8 and 1.2–1.9). Switching at step 4000 moves the stable rank only 0.27 (into Muon) and 0.46 (into AdamW) of the way to the new optimizer's value, and the distance from the initialization 0.49 and 0.70.

Next (Nov–Dec). (i) The equal-loss ranking (Step 2) and its top metrics on the forks (Step 3); (ii) the same on TinyStories-30M and Shakespeare; (iii) presentation (Dec 1–4), paper, code, Colab notebook (Dec 5–11). Code: https://github.com/qqgjyx/MIT-6.7910.

References

Cohen, J. M., B. Ghorbani, S. Krishnan, et al. (2022). *Adaptive Gradient Methods at the Edge of Stability*. Preprint.

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A Hyperparameters

Table 2: TinyStories-10M, shared by the AdamW and Muon runs. Values from the repository's configuration files and code.

Runs	AdamW and Muon × data orders d0–d4 (10)
Model	nanoGPT, pre-LayerNorm blocks
Layers, heads, width	6, 6, 384
MLP	GELU, width 1536
Biases, dropout	none, none
Output head	untied
Attention projections	separate query, key, value
Context	256 tokens
Parameters	13.9M (10.6M without token and position embeddings and head)
Data	TinyStories (GPT-4 version)
Tokenizer	byte-level BPE, 4096 tokens, trained on the training split
Batch	128 windows × 256 tokens
Window offsets	from the training split; generator seeded with (data order, step), same for every optimizer
Steps	10,000
Tokens	328M (about 0.6 epochs)
Learning rate	$\eta\,\varphi(t)$ at step t, every parameter group
Warmup	linear, $\varphi(t) = (t+1)/200$ for $t < 200$
Decay	cosine to $\varphi = 0.1$ at step 10,000
Gradient clipping	none
Initialization seed	0, all runs
Training precision	bfloat16 autocast
Prediction, Hessian precision	float32, no autocast (TF32 matrix products allowed)
Logged predictions	next-token distributions: 128 fixed validation sequences × 16 evenly spaced positions (2,048), as float16 log-probabilities
Loss for matching runs	64 fixed training sequences, all positions
Both logged at	every 250th step + 30 log-spaced steps in $[1, 10{,}000]$
Hessian	top eigenvalue, 20 Lanczos iterations, 32 fixed training sequences; AdamW also of $P^{-1/2}HP^{-1/2}$
Measured at	every 500th step + 30 log-spaced steps
Loss barrier	64 fixed training sequences, 11 evenly spaced points of the segment between the final weights

Table 3: Per optimizer, TinyStories-10M runs.

	AdamW	Muon
η	$2 \cdot 10^{-3}$	0.01
η choice	best final validation loss of a 5-rate sweep on d0	as AdamW
Update	bias-corrected moments	Nesterov momentum; 5 quintic Newton–Schulz steps in bfloat16
Momentum	$\beta = (0.9, 0.95)$	$\beta = 0.95$
ϵ	10^{-8}	–
Update scale	–	$\sqrt{\max(1, n_{\text{out}}/n_{\text{in}})}$
Weight decay	0.1, decoupled; none on LayerNorm gains	none
Parameters	all	query, key, value, attention-output and both MLP matrices of every block
Aux. AdamW	–	embeddings, head, LayerNorm gains
Aux. η, β, ϵ	–	$2 \cdot 10^{-3}$, $(0.9, 0.95)$, 10^{-8}
Aux. weight decay	–	none

Table 4: Shakespeare-char and TinyStories-30M: rows that differ from TinyStories-10M; all else as in Tables 2 and 3. "Aux.": Muon's auxiliary AdamW.

	Shakespeare-char	TinyStories-30M
Runs	AdamW, Muon × d0–d4 (10)	AdamW, Muon × d0–d4 (10)
Data	Shakespeare, nanoGPT's 90/10 split	TinyStories
Vocabulary	65 characters	4096 (BPE)
Layers, heads, width	6, 6, 384	8, 8, 512
Parameters (without embeddings, head)	10.8M (10.6M)	29.5M (25.2M)
Batch	64×256	128×256
Steps	1,500	20,000
Tokens	24.6M (about 24 passes over the 1M training characters)	655M (about 1.2 epochs)
Warmup steps	100	400
Cosine decay to step	1,500	20,000
η_{AdamW}	$3 \cdot 10^{-4}$	$2 \cdot 10^{-3}$
η_{Muon} (aux.)	0.04 (10^{-3})	0.01 ($1.5 \cdot 10^{-3}$)
η chosen by	minimum validation loss over training (runs overfit)	best final validation loss
AdamW β	$(0.9, 0.99)$, as nanoGPT	$(0.9, 0.95)$
Predictions	all 256 positions of 64 validation sequences (16,384)	16 positions of 128 sequences (2,048)
Predictions at	every 50th + 40 log-spaced	every 500th + 30 log-spaced

B Swaps

Table 5: Swap forks on TinyStories-10M. A: parent's optimizer; B: optimizer after the swap; s: fork step.

Parents	the d0 runs of AdamW and Muon (Table 2)
Shadow state	the other optimizer, stepped on the same mini-batch gradients with learning rate 0: momenta, second moments and step counts evolve as if training; parameters follow A alone
Fork steps s	50, 100, 200, 400, 700, 1000, 1500, 2500, 4000
Saved at s	parameters; states of both optimizers
Swap $A \to B$ loads	parameters at s; B's shadow state
Hyperparameters	B's (Table 3)
Steps	s to 10,000, on the parent's schedule and batches (data order 0)
Swap forks	2 directions × 9 steps = 18
Same-batch control	the parent itself (A continued with the same batches)
Data-order control	A continued from s with its own state, data order 1: 18 forks

C Compute

All runs used one NVIDIA GPU each on the MIT ORCD cluster. GPU-hours are those Slurm allocated to each suite, shared out over the runs of the suite by their training time where runs of several suites shared a GPU. Not counted: the offline measurements (Muon's spectral-norm sharpness, loss barriers, distances), a few GPU-hours.

Table 6: Runs described in the checkpoint and their compute.

	Runs	GPU-hours
Learning-rate sweeps, AdamW and Muon (3 settings)	30	7.1
Runs from the initialization, AdamW and Muon (3 settings, 5 data orders)	30	11.6
Swaps and data-order forks (Table 5)	36	7.1
Step 2 (Table 1): rate search, runs at the chosen rate, runs at a constant rate	$15 + 3 + 16$	8.2
Total	130	34.0
Running: Step 2 runs with consecutive checkpoints (3 optimizers × 3 data orders)	9	–
Running: swaps again, with checkpoints at steps 9998–10,000	18	–